

A reflexive smooth support space preserving weak convergence

Solution packet for arXiv:1505.06235

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Status: claimed full solution; likely valid. Every weakly convergent sequence of Borel probability measures on a separable Banach space can be realized on one separable reflexive Banach subspace that is compactly embedded in the original space, has an equivalent Fréchet-differentiable norm, carries all the measures, and preserves their weak convergence.

1 Source question

Ostrovsky and Sirota prove that weak convergence of probability measures on a separable Banach space X can be preserved after passing to a separable Banach subspace compactly embedded in X [1]. In concluding remark C on page 8, they recall Buldygin's theorem that a probability measure may be concentrated on such a subspace which is additionally reflexive and has a Fréchet-differentiable norm, and ask whether one can retain these "good" properties while also preserving the weak convergence. The source passage is reproduced below.

C. Open question.

V.V.Buldygin in [4] proved that in the probabilistic case $\mu(X) = \mu_n(X) = 1$ the single subspace $(Y, \|\cdot\|)$ may be constructed to be reflexive and with continuous differentiable on the unit sphere in the Fréchet sense norm.

We do not know either or not possible to choose this "good" subspace $(Y, \|\cdot\|)$ such that all the r.v. are concentrate in this space and in addition weakly converges therein.

In precise terms: if $\mu_n \Rightarrow \mu$ on X , can one choose a single separable reflexive Banach space $Y \subset X$, compactly embedded in X and Fréchet-smoothly renormable, such that every μ_n and μ is concentrated on Y and $\mu_n \Rightarrow \mu$ also as measures on Y ? The answer is yes.

2 Input theorem and proof intuition

We use the following published support theorem of Buldygin, in exactly the form recalled in the source paper [2]: every Borel probability measure on a separable Banach space B is concentrated on the range of a compact continuous injection $j : R \rightarrow B$, where R is a separable reflexive Banach space. The smooth norm will be recovered at the end by equivalent renorming.

The main difficulty is that putting all measures on one reflexive support does not itself make the family tight in the stronger support norm. We avoid this by using a Skorokhod coupling $\xi_n \rightarrow \xi$ almost surely and treating the whole difference sequence

$$D = (\xi_n - \xi)_{n \geq 1}$$

as one random variable with values in $c_0(X)$. Buldygin's theorem puts D on a reflexive space Z compactly embedded in $c_0(X)$. Compactness makes the coordinate operators $Z \rightarrow X$ uniformly small in the tail.

Those coordinate ranges still need to be assembled inside one reflexive subspace of X . We group them into finite blocks, take an ℓ_1 -sum inside each block, and an ℓ_2 -sum across blocks. Increasing block weights make the norm of the n th coordinate tend to zero, while sufficiently small coordinate-operator tails make the assembled operator compact. This is the only new mechanism in the proof.

3 Compact coordinate assembly

For $n \geq 1$, let $Q_n : c_0(X) \rightarrow X$ be the n th coordinate map.

Lemma 1 (reflexive coordinate assembly). *Let Z be a separable reflexive Banach space and $J : Z \rightarrow c_0(X)$ a compact operator. Set $A_n = Q_n J$. Then there are a separable reflexive Banach space $\mathcal{E}$ and a compact operator $T : \mathcal{E} \rightarrow X$ such that*

$$A_n z \in T(\mathcal{E}) \quad \text{and} \quad \|A_n z\|_{T(\mathcal{E})} \longrightarrow 0 \quad (z \in Z),$$

where the range is equipped with the quotient norm induced by T .

Proof. Compact subsets of $c_0(X)$ have uniformly vanishing coordinate tails. Indeed, this follows by taking a finite uniform-norm net and then taking a common tail for its finitely many centers. Applied to the closure of $J(B_Z)$, this gives

$$\sup_{n \geq N} \|A_n\| \longrightarrow 0. \tag{1}$$

Choose integers

$$1 = N_1 < N_2 < N_3 < \dots$$

so that, for $k \geq 2$,

$$\sup_{n \geq N_k} \|A_n\| \leq 4^{-k}. \tag{2}$$

Put $I_k = \{N_k, \dots, N_{k+1} - 1\}$ and

$$E_k = \ell_1(I_k; Z), \quad \mathcal{E} = \left(\bigoplus_{k \geq 1} E_k \right)_{\ell_2}.$$

Each I_k is finite, hence each E_k is reflexive; consequently $\mathcal{E}$ is reflexive and separable. Define

$$T((z_n)_{n \geq 1}) = \sum_{k \geq 1} 2^k \sum_{n \in I_k} A_n z_n. \tag{3}$$

The norm of the k th block map is at most

$$\beta_k := 2^k \sup_{n \in I_k} \|A_n\|.$$

Here $\beta_1 < \infty$ and, by (2), $\beta_k \leq 2^{-k}$ for $k \geq 2$. Thus $(\beta_k) \in \ell_2$, and Cauchy–Schwarz shows that the series in (3) converges and defines a bounded operator.

Each $A_n = Q_n J$ is compact. Therefore every finite block truncation of T is compact, while the operator norm of the omitted tail is at most

$$\left(\sum_{k > K} \beta_k^2 \right)^{1/2} \longrightarrow 0.$$

Hence T is compact.

If $n \in I_k$ and $z \in Z$, place $2^{-k}z$ in the n th coordinate of E_k and zero elsewhere. Formula (3) maps this vector to $A_n z$, and its $\mathcal{E}$ -norm is $2^{-k}\|z\|_Z$. Consequently

$$\|A_n z\|_{T(\mathcal{E})} \leq 2^{-k}\|z\|_Z \quad (n \in I_k), \quad (4)$$

which tends to zero with n . □

4 Full affirmative answer

Theorem 1. *Let X be a separable Banach space and let μ_n, μ be Borel probability measures on X such that $\mu_n \Rightarrow \mu$ weakly. There exists a separable reflexive Banach space $Y \subset X$ with the following properties:*

1. *the inclusion $Y \hookrightarrow X$ is compact;*
2. *$\mu_n(X \setminus Y) = \mu(X \setminus Y) = 0$ for every n ;*
3. *after the measures are regarded as measures on Y , $\mu_n \Rightarrow \mu$ weakly on Y ;*
4. *Y admits an equivalent norm which is Fréchet differentiable on its unit sphere.*

Thus the open question in concluding remark C of [1] has an affirmative answer.

Proof. Since X is Polish, the Skorokhod representation theorem gives X -valued random variables ξ_n, ξ on one probability space, with laws μ_n, μ , such that

$$\xi_n \longrightarrow \xi \quad \text{almost surely in } X. \quad (5)$$

After changing the variables on a null set, define the $c_0(X)$ -valued random variable

$$D = (d_n)_{n \geq 1}, \quad d_n = \xi_n - \xi.$$

It is Borel measurable because its finite coordinate truncations are measurable and converge to it in the $c_0(X)$ norm.

Apply Buldygin's support theorem twice. For the law of ξ there are a separable reflexive space R and a compact injection $j_0 : R \rightarrow X$ such that

$$\xi \in j_0(R) \quad \text{almost surely.} \quad (6)$$

For the law of D there are a separable reflexive space Z and a compact injection $J : Z \rightarrow c_0(X)$ such that

$$D \in J(Z) \quad \text{almost surely.} \quad (7)$$

Apply Lemma 1 to J , obtaining $\mathcal{E}$ and T . Let

$$\mathcal{H} = R \oplus_2 \mathcal{E}, \quad S : \mathcal{H} \rightarrow X, \quad S(r, e) = j_0 r + T e.$$

The space $\mathcal{H}$ is separable and reflexive, and S is compact. Set

$$Y = \mathcal{H} / \ker S$$

and identify Y with $S(\mathcal{H}) \subset X$, equipped with the quotient norm. Then Y is separable and reflexive, and its inclusion into X is compact.

On the probability-one event where (6) and (7) hold, write $\xi = j_0 r$ and $D = Jz$. Since $d_n = Q_n Jz = A_n z$, Lemma 1 shows that $d_n \in Y$ and

$$\|d_n\|_Y \longrightarrow 0.$$

Thus $\xi, \xi_n = \xi + d_n$ all belong to Y almost surely and

$$\xi_n \longrightarrow \xi \quad \text{almost surely in } Y. \quad (8)$$

The continuous injection of the separable Banach space Y into the Polish space X has Borel range and Borel inverse on its range by the Lusin–Souslin theorem. Hence the variables above are genuinely measurable as Y -valued random variables, and their laws are precisely the unique lifts of μ_n, μ to Y . For every bounded continuous $f : Y \rightarrow \mathbb{R}$, (8) and dominated convergence give

$$\int_Y f d\mu_n \longrightarrow \int_Y f d\mu.$$

This proves weak convergence on Y .

Finally, a separable reflexive Banach space has separable dual, and every Banach space with separable dual admits an equivalent norm which is Fréchet differentiable away from the origin [3]. Replace the quotient norm of Y by such an equivalent norm. Equivalent renorming preserves compactness of the inclusion, concentration, and convergence, and gives property 4. $\square$

5 Verification, scope, and novelty

The proof separates the external inputs from the new argument. Its external inputs are the Skorokhod representation theorem, Buldygin’s single-measure reflexive-support theorem, Lusin–Souslin measurability, and the classical smooth-renorming theorem. The coordinate-assembly lemma is proved in full. The critical estimates are (2)–(4); they simultaneously give compactness of T and convergence of every coordinate in the new range norm. No computational step is used.

A bounded novelty search on 22 July 2026 covered arXiv and web searches for arXiv:1505.06235, the exact paper title and open-question wording, Buldygin’s support theorem and DOI, and combinations of “compactly embedded,” “reflexive subspace,” “Fréchet differentiable norm,” and “weak convergence of probability/Radon measures.” It found the source paper, Buldygin’s support result, and related later work with additional moment or uniform-integrability assumptions, but no later result explicitly resolving this exact question. Novelty confidence is *moderate*; exhaustive priority is not claimed.

The theorem answers the source’s probabilistic question for sequences. It does not claim the analogous statement for the source’s arbitrary nets of sigma-finite measures. Recommended human review: **verify and circulate**, focusing on the interpretation of Buldygin’s theorem, the compact block assembly, and the Borel lifting step.

References

- [1] E. Ostrovsky and L. Sirota, *Strengthening of weak convergence for Radon measures in separable Banach spaces*, 2015, arXiv:1505.06235.
- [2] V. V. Buldygin, *The supports of probability measures in separable Banach spaces*, Theory Probab. Appl. **29** (1985), 546–549; doi:10.1137/1129067.
- [3] R. Deville, G. Godefroy, and V. Zizler, *Smoothness and Renormings in Banach Spaces*, Pitman Monographs and Surveys in Pure and Applied Mathematics 64, Longman Scientific & Technical, 1993.